HW0317

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3.18

3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\underset{(\text{se})}{\widehat{INSURANCE}} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880}INCOME$$

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

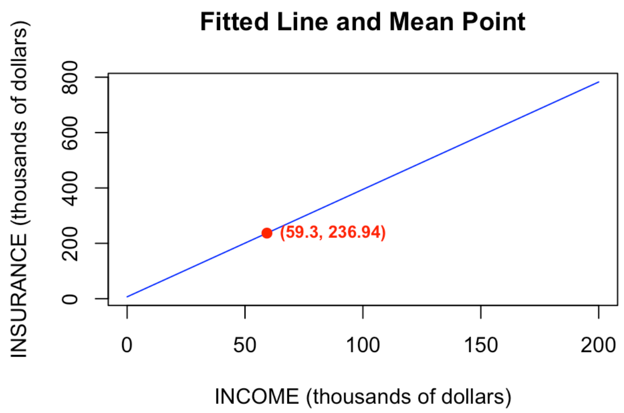
b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is −0.746.

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

(a.)

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根據估計式：INSURANCE_hat = 6.855 + 3.88 * INCOME
當 INCOME 的平均值為： 59.3
計算得出 INSURANCE 的平均值為： 236.939
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Mean_INSURANCE=236.94

(b.)

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點估計值為： 3.88
95% 信賴區間為：[ 3.644697 , 4.115303 ]
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Point Estimate = 3.880 ; 95% Interval Estimate = [3.645, 4.115]

Explain:

The probability that the average amount of insurance held changing with each additional $1000 of household income falls within [$3,645, $4,115] is 95%.

(c.)

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99% 信賴區間為：[ 378.8936 , 410.8164 ]
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99% Interval Estimate = [378.8936, 410.8164]

(d.)

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虛無假設 H0: beta2 = 5 對立假設 H1: beta2 ≠ 5

拒絕域為絕對值t大於 2.100922 t值為 -10

結論：拒絕虛無假說 (Reject H0)。數據「不支持」管理層的說法。
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$H_0: \beta_2 = 5$

$H_1: \beta_2 \neq 5$

t statistic = -10

Rejection region: $|t|>t_c=2.101$

The result doesn't support the claims because the t statistic falls in the rejection region.

(e.)

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虛無假設 H0: beta2 = 1 對立假設 H1: beta2 > 1
拒絕域為 t 大於 2.55238 t 值為 25.71429
結論：拒絕虛無假說 (Reject H0)。數據顯示保險持有量的增加幅度「顯著大於」收入的增加幅度。
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$H_0: \beta_2 = 1$

$H_1: \beta_2 > 1$

t statistic = 25.714

Rejection region: $t>t_c=2.552$

We reject H_0 and conclude that there is a statistically significant that the amount of life insurance held increases by a greater amount as income increases.

3.23

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in $1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SOFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to $13000 against the alternative that the marginal effect will be greater than $13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to $13000 against the alternative that the marginal effect will be greater than $13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

(a.)

虛無假設 H0: me <= 13 對立假設 H1: me > 13

t 值為 -26.72875 拒絕域為 t 大於 1.647919 p-value: 1

結論：無法拒絕虛無假說 (Fail to reject H0)。

t statistic = -26.72875

Rejection region: $t > t_c = 1.647919$

p-value=1

We fail to find sufficient evidence to conclude that the marginal effect of a 2000-square-foot house is significantly greater than $13,000.

(b.)

虛無假設 H0: me <= 13 對立假設 H1: me > 13

t 值為 4.189467 拒絕域為 t 大於 1.647919

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p-value: 1.65395e-05
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結論：拒絕虛無假說 (Reject H0)。數據顯示4000 平方英尺房子的邊際效應超過 $13,000。
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t statistic = 4.189467

Rejection region: $t > t_c = 1.647919$

p-value= 0.0000165395

We reject H_0 and conclude that the marginal effect of a 4000-square-foot house is significantly greater than $13,000.

(c.)

```
       fit      lwr      upr
1 167.3735 158.0481 176.6988
```

95% Interval Estimate = [158.0481, 176.6988]

Explain:

The probability that the price for a house of 2000 square feet falls within [$158,048, $176,699] is 95%.

(d.)

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[1] 138 169 183 樣本中 SQFT=20 的平均房價為： 163.3333
```

In the sample, there are 3 houses with 2000 square feet. They sold for $138,000, $169,000

and $183,000. The average is 163.3333 which is inside the interval estimate.

Since the actual sample mean of 163.3333 falls within this interval, the observed data is consistent with the model’s prediction.

3.31

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable $SAL1$ = unit sales of brand no. 1 canned tuna, and $APR1$ = price per can of brand no. 1 tuna (in dollars).

a. Calculate the summary statistics for $SAL1$ and $APR1$. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus $WEEK$. How much variation in sales and price is there from week to week?

b. Plot the variable $SAL1$ (y-axis) against $APR1$ (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?

c. Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

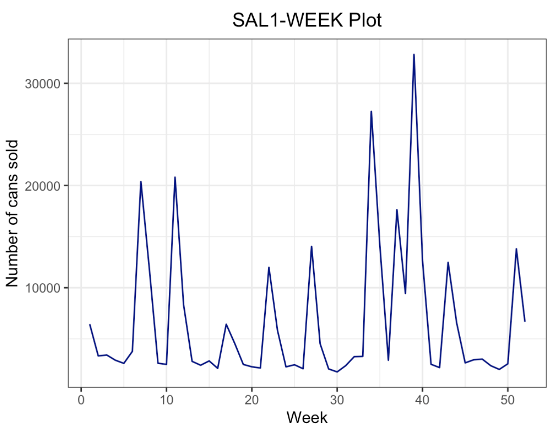
d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

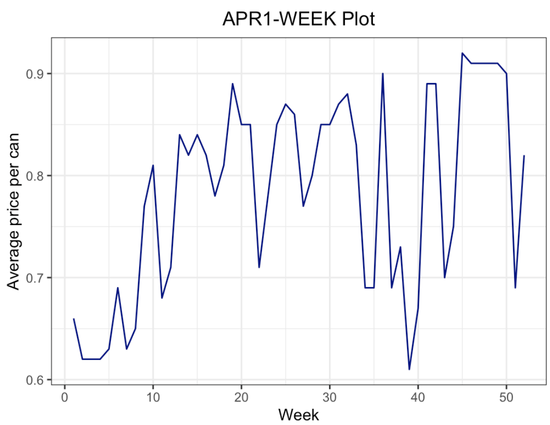
e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

(a.)

	Variable	Mean	Min	Max	SD
1	SAL1	6718.7115	1762.00	32820.00	6.915083e+03
2	APR1	0.7825	0.61	0.92	9.803711e-02

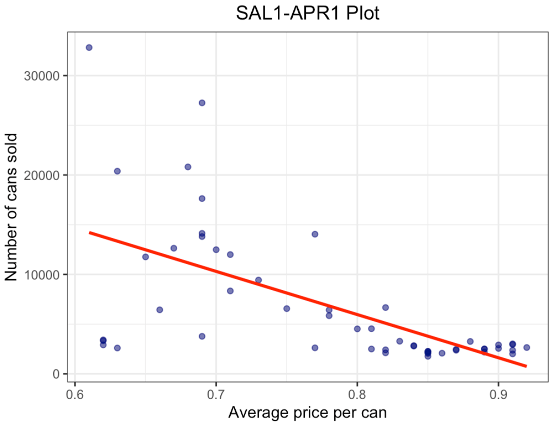




銷量的 CV: 1.029228

價格的 CV: 0.125287

(b.)



Inverse relationship

Yes.This is because that if the price get higher, the sales volume will be lower for the perspective of demand curve.

(c.)

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Call:
lm(formula = sal1 ~ price1, data = tuna)

Residuals:
     Min       1Q   Median       3Q      Max
-10869.5  -1588.3   -499.3   1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
price1       -434.45      78.58  -5.528 1.17e-06 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,	Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

Point Estimate = -434.45

```
            2.5 %    97.5 %
price1 -592.2897 -276.6049
```

<— 95% interval estimate

(d.)

```
      fit      lwr      upr
1 10302.9 8624.934 11980.87
```

90% interval estimate:[8624.934, 11980.87]

(e.)

價格彈性估計結果
點估計值 (Point Estimate): -5.059825
95% 信賴區間 (95% CI): [-6.898149 , -3.221501]

Fairly Elastic (The absolute value of the 95% interval estimate is totally larger than 1.)

Yes. Due to so many substitutes of the tuna cans, the sales volume will be lower numerously if the price get a little bit higher.

(f.)

虛無假設 H0: elasticity = -3 對立假設 H1: elasticity ≠ -3

t 值為 -2.250572 拒絕域為絕對值 t 大於 1.675905

p-value: 0.02884204

結論：拒絕虛無假說 (Reject H0)。數據支持彈性係數應不等於3。

We reject H_0 and conclude that the price elasticity is significantly unequal to 3.